

Analysis of Algorithm

what is the time complexity of the given function? (3 p)

```
def example4(S):  
    n = len(S)  
    prefix = 0  
    total = 0  
    for j in range(n):  
        prefix += S[j]  
        total += prefix  
    return total
```

- A. $O(n^3)$
- B. $O(n)$
- C. $O(\log n)$
- D. $O(n^2)$

Solution: B

////////////////////////////////////

what is the time complexity of the given function? (3 p)

```
def ex4( n ):  
    count = 0  
    for i in range( n ) :  
        for j in range( 25 ) :  
            count += 1  
    return count
```

- A. $O(n^3)$
- B. $O(n)$
- C. $O(\log n)$
- D. $O(n^2)$

Solution: B

////////////////////////////////////

what is the time complexity of the given function? (3 p)

```
sum = 0
while n > 0 :
    sum += n % 10
    n //= 10
```

- A. $O(n)$
- B. $O(\log n)$
- C. $O(n \log n)$
- D. $O(n^2)$

Solution: B

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what is the time complexity of the given function? (3 p)

```
j = n
while j >= 1:
    for i in range(1, n + 1):
        x = x + 1
    j = j // 2
```

- A. $O(n)$
- B. $O(\log n)$
- C. $O(n^2)$
- D. $O(n \log n)$

Solution: D

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Arrays and Lists

Given an array `arr[0.....9][0.....14]` with a base value of 100 and the size of each element is 1 Byte in memory find the address of `arr[8][6]` with the help of **column-major** order. (4 p)

- A. 157
- B. 237
- C. 226
- D. 168

Solution: D

Given:

Base address $B = 100$

Storage size of one element store in any array $W = 1$ Bytes

Row Subset of an element whose address to be found $I = 8$

Column Subset of an element whose address to be found $J = 6$

Lower Limit of row/start row index of matrix $LR = 0$

Lower Limit of column/start column index of matrix $= 0$

Number of column given in the matrix $M = \text{Upper Bound} - \text{Lower Bound} + 1$
 $= 9 - 0 + 1$
 $= 10$

Formula: used

Address of $A[I][J] = B + W * ((J - LC) * M + (I - LR))$

Address of $A[8][6] = 100 + 1 * ((6 - 0) * 10 + (8 - 0))$

Address of $A[I][J] = 168$

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Stack and Queue

You are given a **Stack** and numbers 1,3,4,5,7 respectively. The order of entering number is from left to right.

you apply **push** and **pop** respectively. Which one of the following outputs from this stack is not possible.? (6 p)

- A. 7,5,4,3,1
- B. 1,3,7,5,4
- C. 1,7,3,5,4
- D. 1,4,3,7,5

Solution: C

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You are given two queues. (6 p)

Q1: 10, 25,17,41,19,26,75

Q2: 1,5,7,4,9,6

If X and Y are the elements of the queue, then what would be the elements in Q3 after running this code?

```
MakeNull(Q3);  
i=0;  
while( not empty(Q1) and not empty(Q2))  
    i=i+1;  
    x=DeleteQ(Q1);  
    y=DeleteQ(Q2);  
    if (y=i) then Add(Q3,x)  
end while
```

- A. Q3: 1,4,6
- B. Q3: 10,25,17
- C. Q3: 10,41,26
- D. Q3: 1,5,7

Solution: C

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you are given a python code to insert and delete elements in a circular queue. Which of the options should be placed in the code to complete the program and to delete the first element from the circular queue ? (6 p)

```

class MyCircularQueue():

    def __init__(self, k):
        self.k = k
        self.queue = [None] * k #k is the size of queue
        self.front = self.rear = -1

    # Insert an element into the circular queue
    def enqueue(self, data):

        if ((self.rear + 1) % self.k == self.front):
            print("The circular queue is full\n")

        elif (self.front == -1):
            self.front = 0
            self.rear = 0
            self.queue[self.rear] = data
        else:
            self.rear = (self.rear + 1) % self.k
            self.queue[self.rear] = data

    # Delete an element from the circular queue
    def dequeue(self):
        if (self.front == -1):
            print("The circular queue is empty\n")

        elif (self.front == self.rear):
            temp = self.queue[self.front]
            self.front = -1
            self.rear = -1
            return temp
        else:
            temp = self.queue[self.front]

            */-----HERE-----/*

            return temp

```

- A. self.front = self.front - 1
- B. self.front = (self.front + 1) % self.k
- C. self.front = self.front + 1

D. `self.front = (self.front - 1) % self.k`

Solution: B

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Linked list

given code creates linked list by adding node 6 into an empty list. And then add the other nodes into the list. (6 p)

Fill in the blank.

```
# Represents node of the linked list
class Node:
    def __init__(self, data):
        self.data = data

def addToEmptyList(head, data):
    if head:
        return head
    temp = Node(0)
    temp.data = data
    head = temp
    head.next = head
    return head

def addBegin(head, data):
    if head is None:
        return addToEmptyList(head, data)
    temp = Node(0)
    temp.data = data

    /*-----HERE-----*/
    /*-----HERE-----*/

    return head

head = None
head = addToEmptyList(head, 6)
for i in range(5,0,-1):
    print(i)
    head = addBegin(head,i)
```

Solution:

```
temp.next = head.next
```

```
head.next = temp
```

```
////////////////////////////////////
```

following the previous code the given code below traverses the linked list and prints the node values in the output. What is the order of node values printed in the output? (6 p)

```
def traverse(head):
    if head is None:
        print("list is empty")
        return
    p= head
    while True:
        print(p.data, end = "->")
        p =p.next
        if p ==head:
            break

head = None
head = addToEmpty(head, 6)
for i in range(5,0,-1):
    head = addBegin(head,i)

print("list is: ")
traverse(head)
```

Solution:

6->1->2->3->4->5->

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Recursion

in the given recursive function, what would be the result of function with the given inputs

n = 5. (6 p)

```
def f(n):  
    if n == 0 : return 0  
    if n == 1 : return 1  
    else:  
        return f(n-1) + f(n-2) + 1
```

- A. 8
- B. 14
- C. 12
- D. 15

Solution: C

$$f(2) = f(1) + f(0) + 1 = 1 + 0 + 1 = 2$$

$$f(3) = f(2) + f(1) + 1 = 2 + 1 + 1 = 4$$

$$f(4) = f(3) + f(2) + 1 = 4 + 2 + 1 = 7$$

$$f(5) = f(4) + f(3) + 1 = 7 + 4 + 1 = 12$$

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Sorting algorithm

Trace the **Insertion algorithm** over the array here. A = [9, 6, 7, 2, 5, 8]

What would be the array after 5 passes of sorting? (6 p)

Write your answer in the box.

Solution: A = [2, 5, 6, 7, 8, 9]

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Hashing

we have a list of size 20 ($m = 20$). We want to put some elements in Quadratic Probing fashion. The elements are {96, 48, 63, 29, 87, 77, 48, 65, 69, 94, 61}

Which element will be stored in the cell number 10 in the hash table.? (6 p)

- A. 69
- B. 48
- C. 29
- D. 63

Solution: A

we have a list of size 20 ($m = 20$). We want to put some elements in linear probing fashion. The elements are {96, 48, 63, 29, 87, 77, 48, 65, 69, 94, 61}

x	$h(x, i) = (h'(x) + i^2) \bmod 20$
96	$i = 0, h(x, 0) = 16$
48	$i = 0, h(x, 0) = 8$
63	$i = 0, h(x, 0) = 3$
29	$i = 0, h(x, 0) = 9$
87	$i = 0, h(x, 0) = 7$
77	$i = 0, h(x, 0) = 17$
48	$i = 0, h(x, 0) = 8$ $i = 1, h(x, 1) = 9$ $i = 2, h(x, 2) = 12$
65	$i = 0, h(x, 0) = 5$
69	$i = 0, h(x, 0) = 9$ $i = 1, h(x, 1) = 10$
94	$i = 0, h(x, 0) = 14$
61	$i = 0, h(x, 0) = 1$

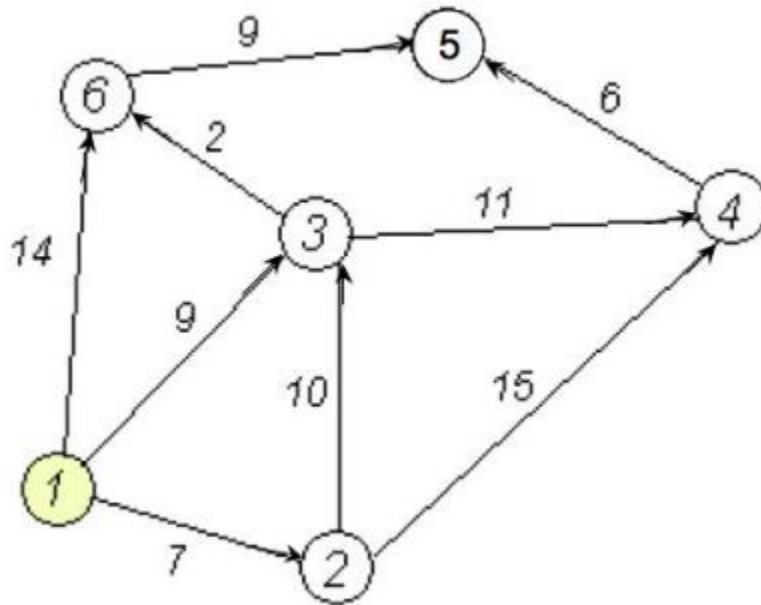
Hash Table

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	61		63		65		87	48	29	69		48		94		96	77		

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Graph

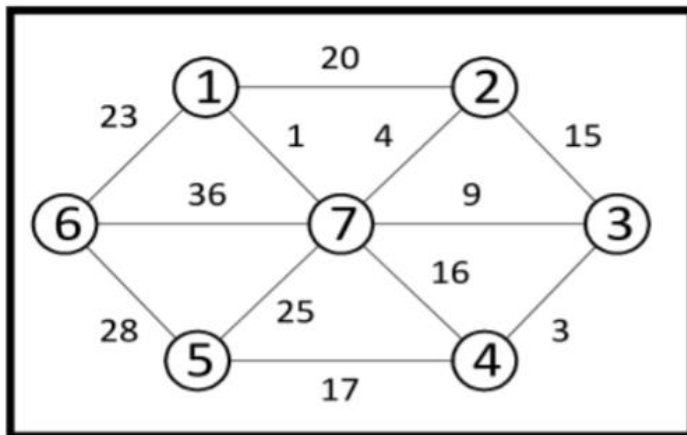
We want to find the shortest path from node 1 to all other nodes using Dijkstra's algorithm. Which option is correct? (6 p)



Solution: 7,9,20,20,11

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what would be the minimum cost for the given spanning tree? (6 p)

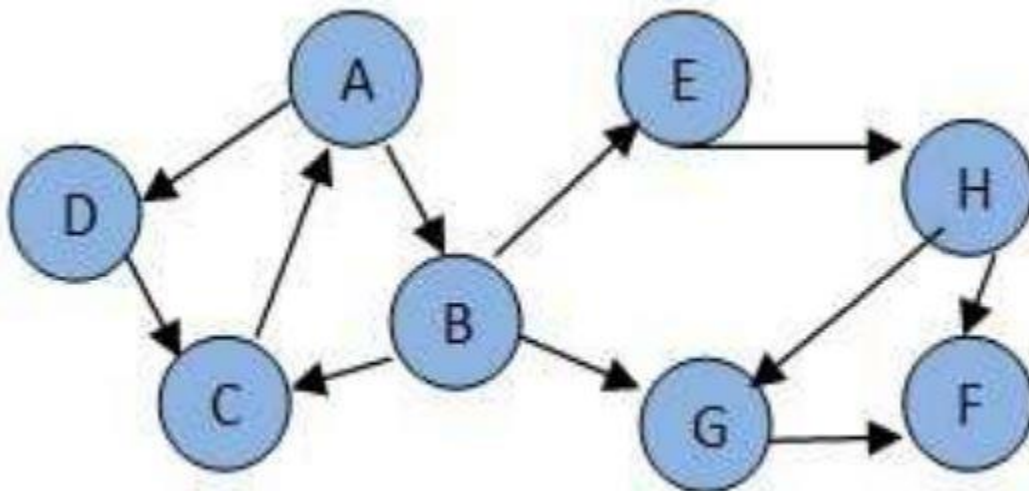


- A. 68
- B. 41
- C. 81
- D. 57

Solution: D

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Starting from node A. In what order will the nodes be visited using a Breadth First Search? (6 p)
assume we always choose the letter closest to the beginning of the alphabet first.



- A. ABDECGHF
- B. ABCDEFGH
- C. ABDCEGHF
- D. ABEDCGHF

Solution: C

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Tree

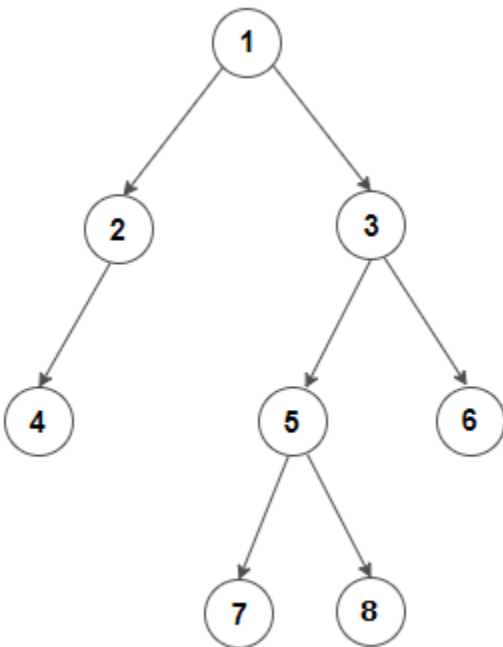
which of the following is a valid pre-order sequence for the given (6 p)

Inn-order = [4, 2, 1, 7, 5, 8, 3, 6]

post-order = [4, 2, 7, 8, 5, 6, 3, 1]

- A. 4, 1, 2, 5, 3, 7, 6, 8
- B. 4, 1, 2, 5, 3, 6, 7, 8
- C. 1, 2, 4, 5, 3, 7, 6, 8
- D. 1, 2, 4, 3, 5, 7, 8, 6

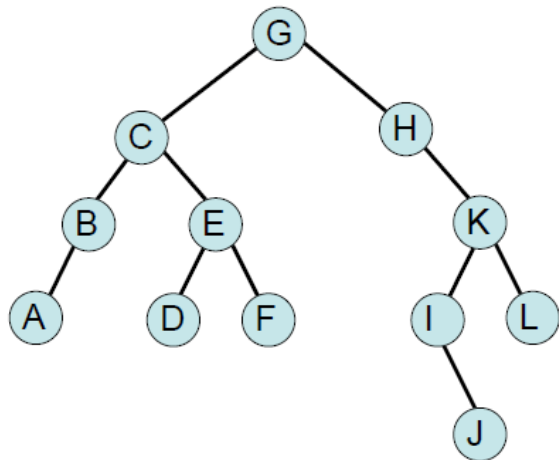
solution: D



The preorder traversal is {1, 2, 4, 3, 5, 7, 8, 6}

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A pre-order traversal of the following tree visits the nodes in which order? (6 p)



- (a) G C B A E D F H K I J L
- (b) A B C D E F G H I J K L
- (c) G C B E A D F H K I L J
- (d) All of these are valid pre-order traversals.

Solution: A

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find the 5th key in the preorder traversal of a Binary Search Tree (BST) with the given post-order? (6 p)

Post-order: 5,6,15,10,23,24,22,26,20

- A. 26
- B. 15
- C. 22
- D. 24

Solution: B

Preorder: 20,10,6,5,15,26,22,24,23

